Introduction to Electronic Design Automation

Homework 2

(Problems 6-8)

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6 Two-level Logic Minimization

6. (a)

Use variable order *abcde*.

$$G = abce + abde + ab\bar{c}e + \bar{a}b\bar{c}e + \bar{a}\bar{b}\bar{c}\bar{e} + \bar{a}\bar{c}\bar{d}e + bc\bar{d}e + c\bar{d}\bar{e}$$

$$D = \bar{a}b\bar{d}e + abc\bar{d} + ac\bar{e}$$

Expanding them into minterms:

$$G = \{0, 1, 2, 4, 9, 11, 12, 13, 20, 25, 27, 28, 29, 31\}$$

$$D = \{9, 13, 20, 22, 28, 29, 30\}$$

Thus, we have the minterms of onset F as:

$$F = G - D = \{0, 1, 2, 4, 11, 12, 25, 27, 31\}$$

Therefore,

$$F \cup D = \{0, 1, 2, 4, 9, 11, 12, 13, 20, 22, 25, 27, 28, 29, 30, 31\}$$

$$\begin{aligned} R &= \{0, 1, ..., 31\} - (F \cup D) \\ &= \{3, 5, 6, 7, 8, 10, 14, 15, 16, 17, 18, 19, 21, 23, 24, 26\} \end{aligned}$$

Writing them as product terms:

$$\begin{aligned} R = {} & \bar{a}\bar{b}\bar{c}de + \bar{a}\bar{b}c\bar{d}e + \bar{a}\bar{b}cd\bar{e} + \bar{a}\bar{b}cde + \bar{a}b\bar{c}\bar{d}\bar{e} + \bar{a}b\bar{c}d\bar{e} \\ & + \bar{a}bcd\bar{e} + \bar{a}bcde + a\bar{b}\bar{c}\bar{d}\bar{e} + a\bar{b}\bar{c}\bar{d}e + a\bar{b}\bar{c}d\bar{e} + a\bar{b}\bar{c}de \\ & + a\bar{b}c\bar{d}e + a\bar{b}cde + ab\bar{c}\bar{d}\bar{e} + ab\bar{c}d\bar{e} \end{aligned}$$

6. (b)

First, we apply the Quine–McCluskey tabular method on $F \cup D$ to generate all prime implicants.

Round 1	Round 2	Round 3
$\bar{a}\bar{b}\bar{c}\bar{d}\bar{e}$ ✓	$\bar{a}\bar{b}\bar{c}\bar{d}$ $\bar{a}\bar{b}\bar{c}\bar{e}$ $\bar{a}\bar{b}\bar{d}\bar{e}$	
$\bar{a}\bar{b}\bar{c}\bar{d}e$ ✓ $\bar{a}\bar{b}\bar{c}d\bar{e}$ ✓ $\bar{a}\bar{b}c\bar{d}\bar{e}$ ✓	$\bar{a}\bar{c}\bar{d}e$ $\bar{b}c\bar{d}\bar{e}$ ✓ $\bar{a}c\bar{d}\bar{e}$ ✓	
$\bar{a}b\bar{c}\bar{d}e$ ✓ $\bar{a}bc\bar{d}\bar{e}$ ✓ $a\bar{b}c\bar{d}\bar{e}$ ✓	$\bar{a}b\bar{c}e$ ✓ $\bar{a}bde$ ✓ $b\bar{c}\bar{d}e$ ✓ $\bar{a}bc\bar{d}$ ✓ $bc\bar{d}\bar{e}$ ✓ $a\bar{b}c\bar{e}$ ✓ $ac\bar{d}\bar{e}$ ✓	$c\bar{d}\bar{e}$ $b\bar{d}e$ $b\bar{c}e$ $bc\bar{d}$ $ac\bar{e}$
$\bar{a}b\bar{c}de$ ✓ $\bar{a}bc\bar{d}e$ ✓ $a\bar{b}cd\bar{e}$ ✓ $ab\bar{c}\bar{d}e$ ✓ $abc\bar{d}\bar{e}$ ✓	$b\bar{c}de$ ✓ $bc\bar{d}e$ ✓ $ac\bar{e}$ ✓ $ab\bar{c}e$ ✓ $ab\bar{d}e$ ✓ $abc\bar{d}$ ✓ $abc\bar{e}$ ✓	abe abc
$ab\bar{c}de$ ✓ $abc\bar{d}e$ ✓ $abcd\bar{e}$ ✓	$abde$ ✓ $abce$ ✓ $abcd$ ✓	
$abcde$ ✓		

Table 1: Quine–McCluskey tabular method for generating prime implicants

Thus, the prime implicants are

$$\left\{\bar{a}\bar{b}\bar{c}\bar{d}, \bar{a}\bar{b}\bar{c}\bar{e}, \bar{a}\bar{b}\bar{d}\bar{e}, \bar{a}\bar{c}\bar{d}e, c\bar{d}\bar{e}, b\bar{d}e, b\bar{c}e, bc\bar{d}, ac\bar{e}, abe, abc\right\}.$$

Next, we use the Column covering of Boolean matrix to find the essential prime implicants. The Boolean matrix is as follows:

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{b}\bar{c}\bar{e}$	$\bar{a}\bar{b}\bar{d}\bar{e}$	$\bar{a}\bar{c}\bar{d}e$	$c\bar{d}\bar{e}$	$b\bar{d}e$	$b\bar{c}e$	$bc\bar{d}$	$ac\bar{e}$	abe	abc
$m_0 = \bar{a}\bar{b}\bar{c}\bar{d}\bar{e}$	1	1	1	0	0	0	0	0	0	0	0
$m_1 = \bar{a}\bar{b}\bar{c}\bar{d}e$	1	0	0	1	0	0	0	0	0	0	0
$m_2 = \bar{a}\bar{b}\bar{c}d\bar{e}$	0	1	0	0	0	0	0	0	0	0	0
$m_4 = \bar{a}\bar{b}c\bar{d}\bar{e}$	0	0	1	0	1	0	0	0	0	0	0
$m_{11} = \bar{a}b\bar{c}de$	0	0	0	0	0	0	1	0	0	0	0
$m_{12} = \bar{a}bc\bar{d}\bar{e}$	0	0	0	0	1	0	0	1	0	0	0
$m_{25} = ab\bar{c}\bar{d}e$	0	0	0	0	0	1	1	0	0	1	0
$m_{27} = ab\bar{c}de$	0	0	0	0	0	0	1	0	0	1	0
$m_{31} = abcde$	0	0	0	0	0	0	0	0	0	1	1

Table 2: Boolean matrix

6. (c)

Rows m_2 and m_{11} each have only one 1, so $\bar{a}\bar{b}\bar{c}\bar{e}$ and $b\bar{c}e$ are essential prime implicants. So we remove the rows m_2, m_{11} and the columns of $\bar{a}\bar{b}\bar{c}\bar{e}$ and $b\bar{c}e$. We also remove the rows that have 1 in the removed columns, which are m_0, m_{25}, m_{27}. After removing these rows, the columns $ac\bar{e}$ and $b\bar{d}e$ contain only zeros, so we remove them as well. The resulting Boolean matrix is as follows:

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{b}\bar{d}\bar{e}$	$\bar{a}\bar{c}\bar{d}e$	$c\bar{d}\bar{e}$	$bc\bar{d}$	abe	abc
$m_1 = \bar{a}\bar{b}\bar{c}\bar{d}e$	1	0	1	0	0	0	0
$m_4 = \bar{a}\bar{b}c\bar{d}\bar{e}$	0	1	0	1	0	0	0
$m_{12} = \bar{a}bc\bar{d}\bar{e}$	0	0	0	1	1	0	0
$m_{31} = abcde$	0	0	0	0	0	1	1

Table 3: Reduced Boolean matrix

6. (d)

Column $c\bar{d}\bar{e}$ covers rows m_4 and m_{12}. It dominates both columns $\bar{a}\bar{b}\bar{d}\bar{e}$ and $bc\bar{d}$, so we remove these two dominated columns. The resulting Boolean matrix is as follows:

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{c}\bar{d}e$	$c\bar{d}\bar{e}$	abe	abc
$m_1 = \bar{a}\bar{b}\bar{c}\bar{d}e$	1	1	0	0	0
$m_4 = \bar{a}\bar{b}c\bar{d}\bar{e}$	0	0	1	0	0
$m_{12} = \bar{a}bc\bar{d}\bar{e}$	0	0	1	0	0
$m_{31} = abcde$	0	0	0	1	1

Table 4: Reduced Boolean matrix

Since rows m_4 and m_{12} have only one 1, the prime implicant $c\bar{d}\bar{e}$ is an induced essential prime implicant. Therefore, we select it and remove rows m_4 and m_{12}.

	$\bar{a}\bar{b}\bar{c}\bar{d}$	$\bar{a}\bar{c}\bar{d}e$	abe	abc
$m_1 = \bar{a}\bar{b}\bar{c}\bar{d}e$	1	1	0	0
$m_{31} = abcde$	0	0	1	1

Table 5: Reduced Boolean matrix

6. (e)

After the reductions in (d), there is no cyclic core to solve, since the remaining matrix is decomposed into two independent single-row choices:

$$m_1 : \{\bar{a}\bar{b}\bar{c}\bar{d}, \bar{a}\bar{c}\bar{d}e\}$$

$$m_{31} : \{abe, abc\}$$

Thus, the independent set heuristic is not applicable. We only need to choose one prime implicant from each set. For example, we choose

$$\bar{a}\bar{b}\bar{c}\bar{d} \quad \text{and} \quad abe.$$

Therefore, the selected prime implicants are

$$\bar{a}\bar{b}\bar{c}\bar{e}, \quad b\bar{c}e, \quad c\bar{d}\bar{e}, \quad \bar{a}\bar{b}\bar{c}\bar{d}, \quad abe.$$

6. (f)

There are four minimum covers with 5 prime implicants:

$$G_1^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{b}\bar{c}\bar{d} + abe. \qquad G_2^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{c}\bar{d}e + abe.$$

$$G_3^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{b}\bar{c}\bar{d} + abc. \qquad G_4^* = \bar{a}\bar{b}\bar{c}\bar{e} + b\bar{c}e + c\bar{d}\bar{e} + \bar{a}\bar{c}\bar{d}e + abc.$$

7 Static Timing Analysis

7. (a)

In this part, we constuct the circuit as a DAG, where each node $v \in V$ represents a gate and each edge $e \in E \subseteq V \times V$ represents a connection between gates. Let $R(v)$ donate the required time at the input of node v, $A(v)$ donate the arrival time at the output of node v, and $d_{u,v}$ donates the delay from node u's output to node v's output. Also, we define the slack of node v as

$$S(v) = R(v) - A(v).$$

Suppose there is a node v such that

$$S(v) = R(v) - A(v) \leq c$$

We want to show that there exists a path satifying every node on the path has slack at most c from some primary input to v and from v to some primary output.

1. From some primary input to v

First, if v is a primary input, then we are done. Otherwise, since

$$A(v) = \max_{u:(u,v)\in E}\big(A(u) + d_{u,v}\big)$$

there exist a node u such that there is an edge from u to v and

$$A(v) = A(u) + d_{u,v}.$$

And since

$$R(u) = \min_{w:(u,w)\in E}\big(R(w) - d_{u,w}\big) \leq R(v) - d_{u,v},$$

we have

$$S(u) = R(u) - A(u) \leq R(v) - d_{u,v} - A(u) = R(v) - A(v) = S(v) \leq c.$$

Thus, we can apply the same argument on u and repeat this process until we reach a primary input.

2. From v to some primary output

First, if v is a primary output, then we are done. Otherwise, since

$$R(v) = \min_{w:(v,w)\in E}\big(R(w) - d_{v,w}\big)$$

there exist a node w such that there is an edge from v to w and

$$R(v) = R(w) - d_{v,w}.$$

And since

$$A(w) = \max_{x:(x,w)\in E} \big(A(x) + d_{x,w}\big) \geq A(v) + d_{v,w},$$

we have

$$S(w) = R(w) - A(w) \leq R(v) + d_{v,w} - A(v) - d_{v,w} = R(v) - A(v) = S(v) \leq c.$$

Thus, we can apply the same argument on w and repeat this process until we reach a primary output.

Hence, there exists a path from some primary input to v and from v to some primary output such that the slack of each node on these paths is at most c.

7. (b)

The equivalence DAG of the giver circuit is as Figure 1 below. The delay of each gate is labled on the corresponding node. Arrival time, required time and slack of each node are also labled on the corresponding node in the format of (a, r, s), where a is the arrival time, r is the required time and s is the slack. The critical path is highlighted in red.

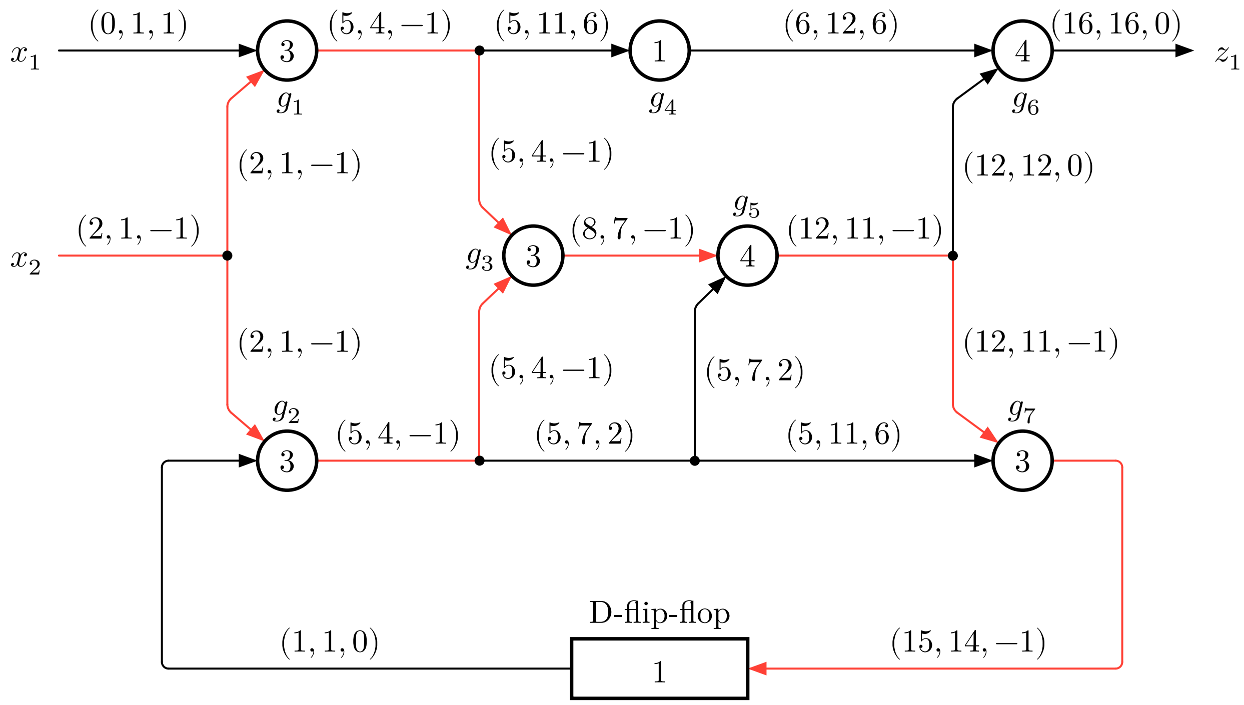

Figure 1: static timing analysis result

8 Technology Mapping